

ON THE KRULL DIMENSIONS OF RINGS OF INTEGER-VALUED POLYNOMIALS AND POLYNOMIAL RINGS

VIET-HOANG TRAN AND TAN M. NGUYEN

ABSTRACT. It was conjectured that every integral domain D satisfies

$$\dim \operatorname{Int}(D) \leq \dim D[X].$$

We disprove this conjecture and, more strongly, classify all pairs for which both dimensions are finite. Writing $m = \dim D[X]$ and $n = \dim \operatorname{Int}(D)$, the pairs that occur are precisely

$$\{(1, 1), (2, 2)\} \cup \{(m, n) \in \mathbb{Z}_{\geq 0}^2 : m \geq 3, n \geq m - 1\}.$$

1. INTRODUCTION

Let D be an integral domain, let $K = \operatorname{Frac}(D)$, and let X be an indeterminate over K . The ring

$$\operatorname{Int}(D) = \{f \in K[X] : f(D) \subseteq D\}$$

lies between $D[X]$ and $K[X]$. The central dimension problem considered in this paper is whether

$$\dim \operatorname{Int}(D) \leq \dim D[X] \quad \text{for every integral domain } D. \quad (1.1)$$

The inclusions $D[X] \subseteq \operatorname{Int}(D) \subseteq K[X]$ do not decide this question, since Krull dimension is not monotone under ring extension.

Fontana and Kabbaj explicitly formulated (1.1) as Conjecture $(\mathcal{C}_1)$ [10]. Chabert later recorded it as a question [5, p. 5], and it was listed as Problem 14 in the collection of Cahen, Fontana, Frisch, and Glaz [4]. We refer to (1.1) as Problem 14.

Cahen proved the universal lower estimate

$$\dim D[X] - 1 \leq \dim \operatorname{Int}(D) \quad (1.2)$$

and exhibited cases of equality [2]. Thus strict inequality in the direction

$$\dim \operatorname{Int}(D) < \dim D[X]$$

was already known: equality in (1.2) gives $\dim \operatorname{Int}(D) = \dim D[X] - 1$. For example, a classical pseudo-valuation-domain construction has $\dim D = 1$, $\dim D[X] = 3$, and $\dim \operatorname{Int}(D) = 2$ [8, Example 1.14(a)]. The unresolved direction was the reverse one, namely, whether $\dim \operatorname{Int}(D) > \dim D[X]$ can occur. Had the conjecture been true, every finite dimension pair

$$(m, n) = (\dim D[X], \dim \operatorname{Int}(D))$$

would have had to lie on one of the two adjacent lines $n = m - 1$ and $n = m$. The standard valuative estimate gives only

$$\dim D[X] - 1 \leq \dim \operatorname{Int}(D) \leq \dim_v(D) + 1, \quad (1.3)$$

see [8, Corollary 1.11], where $\dim_v(D)$ is the supremum of the dimensions of the valuation overrings of D . The right side can be strictly larger than $\dim D[X]$, so (1.3) did not settle (1.1).

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We answer Problem 14 negatively and prove substantially more: we determine the complete set of finite dimension pairs. Here and below all dimensions are Krull dimensions and $\mathbb{Z}_{>0}$ denotes the positive integers.

Theorem 1.1 (Complete finite dimension-pair classification). *For positive integers m, n , the following are equivalent.*

(i) *There is an integral domain D such that*

$$\dim D[X] = m, \quad \dim \operatorname{Int}(D) = n.$$

(ii) *Either $(m, n) = (1, 1)$, or $(m, n) = (2, 2)$, or $m \geq 3$ and $n \geq m - 1$.*

Moreover, for every finite field k and every pair in (ii) with $m \geq 3$, the realizing domain may be chosen to be a local domain $(D_{m,n}, \mathfrak{m}_{m,n})$ satisfying

$$D_{m,n}/\mathfrak{m}_{m,n} \cong k, \quad \dim D_{m,n} = m - 2.$$

The occurrence of the pair $(3, 4)$ answers Problem 14 negatively: some domain satisfies

$$\dim D[X] = 3 < 4 = \dim \operatorname{Int}(D).$$

More strongly, the theorem determines every finite pair: the cases $m = 1, 2$ are rigid, while for every $m \geq 3$ Cahen's lower bound $n \geq m - 1$ is the only restriction. The failure of the proposed upper bound is therefore not exceptional: the excess $\dim \operatorname{Int}(D) - \dim D[X]$ is unbounded. No assertion about pairs involving infinite Krull dimension is intended.

The least possible first coordinate of a counterexample is therefore 3. The decisive construction fixes that coordinate while realizing every prescribed second coordinate $n \geq 3$.

Theorem 1.2 (Base-row realization and counterexamples). *For every finite field $k = \mathbb{F}_q$ and every integer $n \geq 3$, there is a one-dimensional local domain $(D_{3,n}, \mathfrak{m}_{3,n})$ such that*

$$D_{3,n}/\mathfrak{m}_{3,n} \cong k, \quad \dim D_{3,n}[X] = 3, \quad \dim \operatorname{Int}(D_{3,n}) = n.$$

For $n = 3$ the two displayed dimensions agree, whereas every $D_{3,n}$ with $n \geq 4$ violates (1.1). The gap $n - 3$ is arbitrarily large even though $\dim D_{3,n} = 1$ and $\dim D_{3,n}[X] = 3$ are fixed.

The conjecture was supported by substantial positive evidence. Fontana, Izelgue, Kabbaj, and Tartarone proved the desired inequality for finite-dimensional Jaffard domains and obtained formulas for important classes of pseudo-valuation domains [8]; a companion paper treated domains of Krull type [9]. Their 1997 paper produced counterexamples to an analogous comparison for generalized rings $I(B, A)$, but not for $\operatorname{Int}(D)$ itself. Fontana and Kabbaj treated locally essential domains and noted that every example then constructed to test the conjecture satisfied the inequality [10]. Further positive results include the case in which $\operatorname{Int}(D)$ is flat as a $D[X]$ -module [12, Proposition 2.8]. Tamoussit later identified a gap in the original locally essential-domain argument and recovered that result while studying intermediate rings between $D[X]$ and $\operatorname{Int}(D)$ [13]. Further positive results for intermediate rings and for integer-valued polynomials on residually cofinite subsets, under Jaffard and related local hypotheses, were obtained by Chems-Eddin, Ouzaouit, and Tamoussit [7, Theorem 3.14 and Corollaries 3.15–3.16]. These results for special classes remain intact; our counterexamples show that their hypotheses cannot simply be removed.

We briefly place the problem in its broader setting. The prototype

$$\operatorname{Int}(\mathbb{Z}) = \{f \in \mathbb{Q}[X] : f(\mathbb{Z}) \subseteq \mathbb{Z}\}$$

contains $(X^2 - X)/2$ and is strictly larger than $\mathbb{Z}[X]$. The modern theory is commonly traced to Pólya and Ostrowski and became a systematic part of commutative algebra in the 1970s; see the monograph of Cahen and Chabert [3] and Chabert's historical survey [5].

The finite-residue-field case is the source of the dimensional phenomenon. If $D/\mathfrak{m}$ is infinite, Vandermonde interpolation gives

$$(D \setminus \mathfrak{m})^{-1} \text{Int}(D) = D_{\mathfrak{m}}[X].$$

Over $D/\mathfrak{m} \cong \mathbb{F}_q$, by contrast, $X^q - X$ vanishes modulo $\mathfrak{m}$ and can support hidden denominators; Cahen showed that the corresponding fibers may contain chains of arbitrarily prescribed length [2]. Our counterexamples exploit precisely this finite-residue obstruction.

The construction separates the two coordinates. The two-point spectrum of $D_{3,n}$ bounds $\dim D_{3,n}[X]$ by 3, while a residue-transcendental element supplies a prime chain of length 3 and hence equality. By contrast, localization at $X^q - X$ exposes a weak coefficient transform with a prime chain of the required length inside $\text{Int}(D_{3,n})$. A transcendence-degree estimate gives the matching upper bound. Related dimension-transfer results for domains sharing an ideal with finite quotient are classical; see [8, Proposition 1.12] and the extension in [6, Theorem 1.3].

To obtain all remaining pairs, we use a pullback suspension that raises $\dim D$, $\dim D[X]$, and $\dim \text{Int}(D)$ simultaneously by one. Its integer-valued dimension formula is a special case of Tartarone's divided-prime pullback formula, recalled and generalized in [6, Section 1 and Theorem 1.2]. We give a direct proof adapted to our construction. Starting from $D_{3,n-m+3}$ realizes every pair with $3 \leq m \leq n$. Starting instead from the classical pseudo-valuation-domain example $D_{3,2}$ realizes the boundary $n = m - 1$. Low-dimensional rigidity supplies the two remaining pairs $(1, 1)$ and $(2, 2)$.

All rings in the paper are commutative with identity. The length of a finite strict chain of prime ideals is the number of strict inclusions in it.

2. NECESSARY BOUNDS AND LOW-DIMENSIONAL RIGIDITY

This section proves the only-if direction of Theorem 1.1 and shows that no counterexample to Problem 14 can satisfy $\dim D[X] \leq 2$. It also explains why the pair $(2, 1)$, although permitted by (1.2), cannot occur.

Proposition 2.1. *Let D be a domain and suppose that*

$$m = \dim D[X]$$

is finite. Put $n = \dim \text{Int}(D)$, $d = \dim D$, and $v = \dim_v(D)$, allowing n and v initially to be infinite. The following inequalities are interpreted in $\mathbb{Z}_{\geq 0} \cup \{\infty\}$:

$$d + 1 \leq m, \quad d + 1 \leq n, \quad m - 1 \leq n \leq v + 1.$$

Proof. For every finite prime chain $\mathfrak{p}_0 \subsetneq \cdots \subsetneq \mathfrak{p}_\ell$ in D , the chain

$$\mathfrak{p}_0[X] \subsetneq \cdots \subsetneq \mathfrak{p}_\ell[X] \subsetneq (\mathfrak{p}_\ell, X)$$

has length $\ell + 1$ in $D[X]$. Since m is finite, this proves $d \leq m - 1$; in particular d is finite and $d + 1 \leq m$.

For any $a \in D$, evaluation is a surjection

$$\text{ev}_a : \text{Int}(D) \longrightarrow D, \quad f \longmapsto f(a).$$

Its kernel

$$\mathcal{Q}_a = (X - a)K[X] \cap \text{Int}(D)$$

is a nonzero prime, since it contains $X - a$. Taking inverse images of a length- d prime chain in D chosen to begin with (0) , and prepending $(0) \subsetneq \mathcal{Q}_a$, gives $d + 1 \leq n$.

The inequality $m - 1 \leq n$ is Cahen's universal estimate [2, Corollaire 1.2]; see also [8, Corollary 1.11]. For the last inequality, first note that

$$\text{Frac}(\text{Int}(D)) = \text{Frac}(D[X]) = K(X).$$

Thus every valuation overring of $\text{Int}(D)$ is also a valuation overring of $D[X]$. Consequently

$$\dim \text{Int}(D) \leq \dim_v \text{Int}(D) \leq \dim_v(D[X]) = \dim_v(D) + 1;$$

see also [8, 5]. $\square$

Proposition 2.2. *If $\dim D[X] = 1$, then $\dim \text{Int}(D) = 1$. If $\dim D[X] = 2$, then $\dim \text{Int}(D) = 2$.*

Proof. If $\dim D[X] = 1$, Proposition 2.1 gives $\dim D = 0$. A zero-dimensional domain is a field, so $\text{Int}(D) = D[X]$ and both dimensions are 1.

Suppose now that $\dim D[X] = 2$. The same proposition gives $\dim D \leq 1$. The dimension cannot be zero: a zero-dimensional domain is a field, and then $\dim D[X] = 1$. Hence $\dim D = 1$. We claim that $\dim_v(D) = 1$. If not, some valuation overring of D has a chain of primes of length at least two. Localizing that valuation ring at the top of a length-two subchain gives a valuation overring $W \subseteq K$ with

$$(0) \subsetneq Q \subsetneq M, \tag{2.1}$$

where M is the maximal ideal of W .

Every nonzero prime P of a valuation overring of D contracts nontrivially to D . Indeed, if $P \cap D = (0)$, then W_P contains the inverse of every nonzero element of D . Hence

$$K = \text{Frac}(D) \subseteq W_P \subseteq K,$$

so $W_P = K$, contradicting $0 \neq PW_P$. It follows from (2.1) and $\dim D = 1$ that

$$Q \cap D = M \cap D = \mathfrak{p}$$

for a maximal ideal $\mathfrak{p}$ of D .

Choose $\xi \in M \setminus Q$ and write $\xi = a/b$ with $a, b \in D$ and $b \neq 0$. If $b \notin \mathfrak{p}$, then b is a unit of W , whereas

$$a = b\xi \in M \cap D = \mathfrak{p} = Q \cap D.$$

This would imply $\xi \in Q$, a contradiction. Thus $b \in \mathfrak{p} = Q \cap D$. Since Q is an ideal of W , it follows that $a = b\xi \in Q \cap D = \mathfrak{p}$. Hence

$$a, b \in \mathfrak{p}, \quad b \in Q.$$

Consider the upper to zero

$$H = (X - \xi)K[X] \cap D[X] = \ker(D[X] \xrightarrow{X \mapsto \xi} K).$$

It is nonzero because $bX - a \in H$. We next prove

$$H \subseteq \mathfrak{p}[X]. \tag{2.2}$$

Suppose that $f = \sum_i d_i X^i \in H$ has a coefficient outside $\mathfrak{p}$, and let j be the least index with $d_j \notin \mathfrak{p}$. Reduce the equation $f(\xi) = 0$ in the valuation domain W modulo Q . The terms with $i < j$ vanish. Since $\xi \notin Q$, cancellation of $\bar{\xi}^j$ in the domain W/Q gives

$$\bar{d}_j + \sum_{i>j} \bar{d}_i \bar{\xi}^{i-j} = 0.$$

The first term is a unit of the valuation domain W/Q , because $d_j \notin M \cap D$, while the sum belongs to its maximal ideal M/Q , because $\xi \in M$. This is impossible and proves (2.2).

All inclusions in

$$(0) \subsetneq H \subsetneq \mathfrak{p}[X] \subsetneq (\mathfrak{p}, X)$$

are strict prime inclusions: a nonzero constant from $\mathfrak{p}$ is not in H , and $X \notin \mathfrak{p}[X]$. This length-three chain contradicts $\dim D[X] = 2$. Hence $\dim_v(D) = 1$. Finally, Proposition 2.1 yields

$$2 = \dim D + 1 \leq \dim \text{Int}(D) \leq \dim_v(D) + 1 = 2.$$

The conclusion follows. $\square$

We use the following elementary lemma for both base constructions.

Lemma 2.3 (Two-point polynomial criterion). *Let (A, M) be a local domain with quotient field K , residue field k , and $\text{Spec}(A) = \{(0), M\}$. Then $\dim A[X] \leq 3$. Suppose in addition that $A \subseteq V \subseteq K$ and that a homomorphism $\rho : V \rightarrow L$ to a field L satisfies*

$$\ker(\rho|_A) = M, \quad \rho(A) = k.$$

Here $k = A/M$ is identified with the subfield $\rho(A)$ of L . If there are $\theta \in V$ and $0 \neq c \in M$ such that $c\theta \in A$ and $\rho(\theta)$ is transcendental over k , then $\dim A[X] = 3$.

Proof. In a prime chain of $A[X]$, the primes contracting to (0) form an initial block. Localization at $A \setminus \{0\}$ identifies that block, with all strict inclusions preserved, with a chain in the principal ideal domain $K[X]$, so it has at most one edge. The primes contracting to M form the terminal block. Each contains $M[X]$, and quotienting by $M[X]$ identifies that block, again with strict inclusions preserved, with a chain in $k[X]$; it too has at most one edge. There is at most one transition between the blocks, proving $\dim A[X] \leq 3$.

For the second assertion, put

$$Q = (X - \theta)K[X] \cap A[X].$$

This is prime and nonzero because $c(X - \theta) \in Q$. If $f = \sum_i a_i X^i \in Q$, then $\sum_i a_i \theta^i = 0$. Applying ρ gives

$$\sum_i \rho(a_i) \rho(\theta)^i = 0.$$

Since $\rho(a_i) \in k$ and $\rho(\theta)$ is transcendental over k , every a_i lies in M . Hence $Q \subseteq M[X]$, and the inclusion is strict because the nonzero constant c is not in Q . Therefore

$$(0) \subsetneq Q \subsetneq M[X] \subsetneq (M, X)$$

is a prime chain of length 3. □

3. THE COUNTEREXAMPLE FAMILY ON THE BASE ROW

We now construct the family responsible for the failure of Problem 14. Every member will be one-dimensional with polynomial-ring dimension 3, while its integer-valued polynomial dimension is prescribed by n . Fix an integer $n \geq 3$ and a finite field $k = \mathbb{F}_q$. We first build a rank-one valued field with value group $\mathbb{Q}$ and an $(n-2)$ -dimensional local domain as its residue coefficient ring. Put

$$R = k[u_1, \dots, u_{n-2}]_{(u_1, \dots, u_{n-2})}, \quad L = \text{Frac}(R) = k(u_1, \dots, u_{n-2}).$$

For each $\ell \geq 1$, let z_ℓ be transcendental over L . Embed $L(z_\ell)$ into $L(z_{\ell+1})$ by

$$z_\ell \mapsto z_{\ell+1}^{\ell+1}.$$

We henceforth make these identifications, write $t = z_1$ and $z_\ell = t^{1/\ell!}$, and set

$$\Omega = \bigcup_{\ell \geq 1} L(t^{1/\ell!}).$$

On $L(z_\ell)$, define

$$\nu_\ell(h) = \frac{1}{\ell!} \text{ord}_{z_\ell}(h) \quad (0 \neq h \in L(z_\ell)),$$

where ord_{z_ℓ} is the order at $z_\ell = 0$. If $h \in L(z_\ell)$, then

$$\text{ord}_{z_{\ell+1}}(h(z_{\ell+1}^{\ell+1})) = (\ell+1) \text{ord}_{z_\ell}(h).$$

It follows that $\nu_{\ell+1}|_{L(z_\ell)} = \nu_\ell$. The ν_ℓ therefore define a valuation

$$\nu : \Omega^\times \longrightarrow \mathbb{Q}, \quad \nu(t) = 1.$$

Its value group is $\bigcup_{\ell \geq 1} \frac{1}{\ell!} \mathbb{Z} = \mathbb{Q}$. At each finite level, the residue of a rational function regular at $z_\ell = 0$ is its value at zero, so the residue field is L ; taking the directed union does not add any residues.

Let

$$V = \{x \in \Omega : \nu(x) \geq 0\}, \quad \rho : V \longrightarrow L$$

be the valuation ring and its residue map. Thus $\ker \rho = \{x \in \Omega : \nu(x) > 0\}$. If $\gamma \in \mathbb{Q}_{\geq 0}$, choose ℓ so that the denominator of γ divides $\ell!$; this defines a compatible element $t^\gamma \in \Omega$. Set

$$\begin{aligned} S &= R[t^\gamma : \gamma \in \mathbb{Q}_{\geq 0}] \subseteq V, \\ S_+ &= (t^\gamma : \gamma \in \mathbb{Q}_{> 0})S, \\ I &= \{x \in V : \nu(x) \geq 1\}, \\ E &= S + I, \quad P = S_+ + I, \quad D_{3,n} = k + P. \end{aligned} \tag{3.1}$$

We record the elementary structural facts needed below.

Lemma 3.1. *Every $g \in S$ has a unique finite expansion*

$$g = \sum_{\gamma \geq 0} a_\gamma t^\gamma, \quad a_\gamma \in R.$$

Moreover, E is a ring, P is an ideal of E , and

$$\rho(E) = R, \quad \ker(\rho|_E) = P.$$

In particular, $E/P \cong R$, every nonzero element of P has positive ν -value, and $D_{3,n}/P \cong k$.

Proof. Consider a finite L -linear relation among distinct rational powers of t . Choose ℓ so that all exponents become integral multiples of $1/\ell!$. The relation becomes a polynomial relation over L for the transcendental element $t^{1/\ell!}$; hence every coefficient is zero. Thus the rational powers of t are L -linearly independent. Combining equal exponents gives the asserted unique expansion for elements of S .

The set I is an ideal of V , and S_+ is an ideal of S . Therefore

$$(S + I)(S + I) \subseteq S + I, \quad (S + I)(S_+ + I) \subseteq S_+ + I.$$

This proves that E is a ring and that P is an ideal of E .

For a finite expansion $g = \sum a_\gamma t^\gamma$, reduction modulo the maximal ideal of V gives $\rho(g) = a_0$. Hence $\rho(E) = R$ and $P \subseteq \ker(\rho|_E)$. Conversely, suppose $g + z \in E$ has residue zero, where $g \in S$ and $z \in I$. Since $\rho(z) = 0$, the constant coefficient a_0 of g is zero. Thus $g \in S_+$ and $g + z \in P$. This proves $\ker(\rho|_E) = P$. The remaining assertions follow by taking quotients and then restricting ρ to $D_{3,n} = k + P$. Notice here that $k \cap P = (0)$, since nonzero elements of k have value zero while nonzero elements of P have positive value. $\square$

Lemma 3.2. *The ring $D_{3,n}$ is local with maximal ideal $\mathfrak{m}_{3,n} = P$.*

Proof. Since $D_{3,n}/P \cong k$, the ideal P is maximal. Let $a + p \in D_{3,n} \setminus P$, where $a \in k^\times$ and $p \in P$. If $p = 0$, the element is plainly a unit. Otherwise put $\delta = \nu(p) > 0$ and choose $h \geq 0$ with $(h+1)\delta \geq 1$. The finite geometric identity in Ω gives

$$\frac{1}{a+p} = \frac{1}{a} \sum_{j=0}^h \left(-\frac{p}{a}\right)^j + \frac{(-1)^{h+1} p^{h+1}}{a^{h+1}(a+p)}.$$

The finite sum lies in $k + P = D_{3,n}$. Since $\nu(a+p) = 0$, the last summand has value $(h+1)\delta \geq 1$ and consequently lies in $I \subseteq D_{3,n}$. Thus $(a+p)^{-1} \in D_{3,n}$, so every element outside P is a unit. $\square$

We henceforth write $\mathfrak{m}_{3,n} = P$.

Lemma 3.3. *One has*

$$\text{Frac}(D_{3,n}) = \Omega, \quad \text{Spec}(D_{3,n}) = \{(0), \mathfrak{m}_{3,n}\}.$$

In particular, $\dim D_{3,n} = 1$.

Proof. The element t has value 1, so $t \in I \subseteq D_{3,n}$. For every $x \in V$ one has $tx \in I \subseteq D_{3,n}$, and therefore $x = (tx)/t \in \text{Frac}(D_{3,n})$. Hence

$$\Omega = \text{Frac}(V) \subseteq \text{Frac}(D_{3,n}) \subseteq \Omega,$$

which proves $\text{Frac}(D_{3,n}) = \Omega$.

Let $0 \neq f \in \mathfrak{m}_{3,n}$ and let $g \in \mathfrak{m}_{3,n}$. The case $g = 0$ is immediate. If $g \neq 0$, then $\nu(f), \nu(g) > 0$, so there is an integer $N \geq 1$ such that

$$N\nu(g) - \nu(f) \geq 1.$$

It follows that $g^N/f \in I \subseteq D_{3,n}$, and hence $g^N \in fD_{3,n}$. We have proved $\mathfrak{m}_{3,n} \subseteq \sqrt{fD_{3,n}}$. The reverse inclusion follows from $fD_{3,n} \subseteq \mathfrak{m}_{3,n}$ and the primality of $\mathfrak{m}_{3,n}$. Thus

$$\sqrt{fD_{3,n}} = \mathfrak{m}_{3,n} \quad (0 \neq f \in \mathfrak{m}_{3,n}). \quad (3.2)$$

Every nonzero proper prime ideal of $D_{3,n}$ contains such an f , and (3.2) forces that prime to equal $\mathfrak{m}_{3,n}$. Since $D_{3,n}$ is a domain, its only remaining prime is (0) . $\square$

The maximal ideal $\mathfrak{m}_{3,n}$ is not finitely generated. Indeed, if $p_1, \dots, p_\ell$ generated it, after discarding zero generators every element of their ideal would have value at least $\min_i \nu(p_i) > 0$, whereas $t^\varepsilon \in \mathfrak{m}_{3,n}$ for arbitrarily small positive rational ε . This is the source of the coefficient phenomenon below.

4. FIXING THE POLYNOMIAL-RING DIMENSION AT THREE

The next calculation fixes the first coordinate of the counterexample family at its least possible value. Although $D_{3,n}$ is one-dimensional, it is not a Jaffard domain: adjoining one indeterminate raises its dimension by two.

Proposition 4.1. *For the domain in (3.1),*

$$\dim D_{3,n}[X] = 3.$$

Proof. Apply Lemma 2.3 with

$$A = D_{3,n}, \quad M = \mathfrak{m}_{3,n}, \quad K = \Omega, \quad \theta = u_1, \quad c = t.$$

The structural hypotheses follow from Lemmas 3.1 and 3.3. Moreover, $u_1 \in V$, its residue is transcendental over k , and $0 \neq t \in \mathfrak{m}_{3,n}$ with $tu_1 \in S_+ \subseteq D_{3,n}$. The criterion therefore gives $\dim D_{3,n}[X] = 3$. $\square$

5. A WEAK COEFFICIENT TRANSFORM

We next isolate the finite-residue mechanism that lets the second coordinate grow while the first remains fixed. The following localization principle is valid for every local domain with finite residue field and identifies exactly which scalar coefficients become visible after one inverts $X^q - X$.

Proposition 5.1 (Weak coefficient localization). *Let $(T, \mathfrak{n})$ be a local domain with quotient field K_T and residue field $T/\mathfrak{n} \cong \mathbb{F}_q$. Define*

$$\mathcal{C}(T) = \{c \in K_T : \text{there is } N \geq 0 \text{ such that } cy^N \in T \text{ for every } y \in \mathfrak{n}\}. \quad (5.1)$$

Then $\mathcal{C}(T)$ is a subring of K_T . If $F = X^q - X$, then

$$\text{Int}(T) \subseteq \mathcal{C}(T)[X], \quad \text{Int}(T)[1/F] = \mathcal{C}(T)[X, 1/F].$$

Proof. Suppose N_c and N_d witness membership in $\mathcal{C}(T)$ for c and d , respectively. For every $y \in \mathfrak{n}$,

$$\begin{aligned}(c+d)y^{N_c+N_d} &= (cy^{N_c})y^{N_d} + (dy^{N_d})y^{N_c} \in T, \\ cd y^{N_c+N_d} &= (cy^{N_c})(dy^{N_d}) \in T.\end{aligned}$$

The same exponent that works for c also works for $-c$, and $0, 1 \in \mathcal{C}(T)$. Thus $\mathcal{C}(T)$ is a ring.

Let $f = \sum_{j=0}^e c_j X^j \in \text{Int}(T)$. If $e = 0$, then $c_0 = f(0) \in T \subseteq \mathcal{C}(T)$. Assume $e > 0$ and fix $0 \neq y \in \mathfrak{n}$. Evaluate f at

$$y, y^2, \dots, y^{e+1}.$$

These points are distinct: if $1 \leq i < j \leq e+1$, then

$$y^j - y^i = y^i(y^{j-i} - 1) \neq 0,$$

because $y^{j-i} - 1$ is a unit of the local domain T . The determinant of the associated Vandermonde matrix is

$$\Delta_y = \prod_{1 \leq i < j \leq e+1} (y^j - y^i) = y^{M_e} \prod_{1 \leq i < j \leq e+1} (y^{j-i} - 1), \quad (5.2)$$

where

$$M_e = \sum_{i=1}^e i(e+1-i) = \frac{e(e+1)(e+2)}{6}.$$

The values $f(y^i)$ lie in T . Cramer's rule, applied over K_T , gives $\Delta_y c_j \in T$ for every j . The second product on the right of (5.2) is a unit of T , so

$$c_j y^{M_e} \in T \quad (0 \leq j \leq e).$$

The same assertion is automatic when $y = 0$. The integer M_e is independent of y , hence every c_j belongs to $\mathcal{C}(T)$. This proves $\text{Int}(T) \subseteq \mathcal{C}(T)[X]$ and therefore

$$\text{Int}(T)[1/F] \subseteq \mathcal{C}(T)[X, 1/F].$$

Conversely, reduction modulo $\mathfrak{n}$ shows $F(a) = a^q - a \in \mathfrak{n}$ for every $a \in T$. If $c \in \mathcal{C}(T)$ and N witnesses (5.1), then

$$cF(a)^N \in T \quad (a \in T).$$

Thus $cF^N \in \text{Int}(T)$ and $c = (cF^N)/F^N \in \text{Int}(T)[1/F]$. Since $X \in \text{Int}(T)$ and F is invertible in the localization, this yields $\mathcal{C}(T)[X, 1/F] \subseteq \text{Int}(T)[1/F]$ and proves equality. $\square$

We now compute the transform (5.1) for $D_{3,n}$. Write

$$C_{3,n} = \mathcal{C}(D_{3,n}) = \{c \in \Omega : \text{there is } N \geq 0 \text{ such that } cy^N \in D_{3,n} \text{ for every } y \in \mathfrak{m}_{3,n}\}. \quad (5.3)$$

Proposition 5.2 (Coefficient chain). *One has $C_{3,n} \subseteq V$ and $\rho(C_{3,n}) = R$. Put*

$$\mathfrak{q}_0 = \ker(\rho|_{C_{3,n}}), \quad \mathfrak{q}_j = (\rho|_{C_{3,n}})^{-1}((u_1, \dots, u_j)) \quad (1 \leq j \leq n-2).$$

These are prime ideals and form a strict chain

$$(0) \subsetneq \mathfrak{q}_0 \subsetneq \mathfrak{q}_1 \subsetneq \dots \subsetneq \mathfrak{q}_{n-2} \quad (5.4)$$

of length $n-1$, with $C_{3,n}/\mathfrak{q}_{n-2} \cong k$.

Proof. Suppose $c \in \Omega$ has $\nu(c) < 0$. For an arbitrary fixed $N \geq 0$, choose $\varepsilon \in \mathbb{Q}_{>0}$ so small that

$$\nu(c) + N\varepsilon < 0;$$

for $N = 0$ any positive ε works. Then $y = t^\varepsilon \in S_+ \subseteq \mathfrak{m}_{3,n}$, but $cy^N \notin V$ and therefore $cy^N \notin D_{3,n}$. No exponent can witness membership in $C_{3,n}$, so $C_{3,n} \subseteq V$.

Let $c \in C_{3,n}$. The assertion $\rho(c) \in R$ is clear for $c = 0$ or $\nu(c) > 0$. Suppose $\nu(c) = 0$, and choose an exponent N witnessing (5.3). If $N = 0$, then $c \in D_{3,n}$ and $\rho(c) \in k \subseteq R$. If $N > 0$, choose $\varepsilon \in \mathbb{Q}_{>0}$ with $N\varepsilon < 1$ and put

$$d = ct^{N\varepsilon} \in D_{3,n}, \quad \alpha = N\varepsilon.$$

Then $\nu(d) = \alpha \in (0, 1)$, so $d \in P$. Write $d = g + z$ with $g \in S_+$ and $z \in I$. Since $\nu(z) \geq 1 > \alpha = \nu(d)$, one has $\nu(g) = \alpha$. By Lemma 3.1, write

$$g = \sum_{\gamma > 0} a_\gamma t^\gamma, \quad a_\gamma \in R.$$

The least occurring exponent is α . Hence

$$\rho(t^{-\alpha}d) = \rho(t^{-\alpha}g) = a_\alpha \in R,$$

because $\nu(t^{-\alpha}z) \geq 1 - \alpha > 0$. Since $t^{-\alpha}d = c$, this proves $\rho(C_{3,n}) \subseteq R$.

Conversely, take $a \in R$ and $y \in \mathfrak{m}_{3,n} = P$. Since $a \in S \subseteq E$ and P is an ideal of E , one has $ay \in P \subseteq D_{3,n}$. Thus exponent 1 witnesses $a \in C_{3,n}$. The map ρ is the identity on R , so $R \subseteq \rho(C_{3,n})$.

It follows that $C_{3,n}/\mathfrak{q}_0 \cong R$. Taking inverse images of

$$(0) \subsetneq (u_1) \subsetneq \cdots \subsetneq (u_1, \dots, u_{n-2})$$

gives all strict inclusions in (5.4) after $\mathfrak{q}_0$; when $n = 3$, the displayed chain means simply $(0) \subsetneq (u_1)$. The remaining inclusion $(0) \subsetneq \mathfrak{q}_0$ is strict because $0 \neq t \in \mathfrak{q}_0$. The final quotient is $R/(u_1, \dots, u_{n-2}) \cong k$. $\square$

6. COMPUTING THE INTEGER-VALUED POLYNOMIAL DIMENSION

We now complete the decisive dimension calculation for $D_{3,n}$. The coefficient chain gives the lower bound n , while the following field-theoretic estimate gives the matching upper bound. It does not require a finite-generation hypothesis.

Lemma 6.1. *Let k be a field and let T be a domain containing k . If $M = \text{Frac}(T)$ has finite transcendence degree over k , then*

$$\dim T \leq \text{trdeg}_k M. \quad (6.1)$$

Proof. We recall the valuation argument. Krull's valuation theorem says that every finite chain of prime ideals of a domain is dominated by a chain of the same length in a valuation overring of its quotient field; equivalently, $\dim T \leq \dim_v T$ [1, Chapter VI] (see also [11]). One obtains the chain form by successively choosing valuations that dominate the relevant local quotient domains and composing them through their residue fields. Let W be such a valuation overring, with valuation

$$\omega : M^\times \rightarrow \Gamma.$$

Because $k \subseteq W$ and $k^\times \subseteq W^\times$, the valuation is trivial on k .

Suppose W contains a strict chain of $\ell + 1$ primes. Prime ideals of a valuation ring correspond, in reverse order, to convex subgroups of its value group. Thus Γ has a strict chain of $\ell + 1$ convex subgroups. Choose $\gamma_1, \dots, \gamma_\ell \in \Gamma$ successively from the differences of adjacent subgroups. These elements are $\mathbb{Q}$ -linearly independent. To see this, in a nontrivial integral relation take the term belonging to the largest convex layer. All lower-layer terms lie in the preceding convex subgroup, whereas a nonzero multiple of the chosen largest-layer element does not: quotients by convex subgroups are ordered and hence torsion-free. This is a contradiction.

Choose $x_i \in M^\times$ with $\omega(x_i) = \gamma_i$. Distinct monomials in $x_1, \dots, x_\ell$ have distinct values, by the $\mathbb{Q}$ -linear independence of the γ_i . Since coefficients from $k^\times$ have value zero, a nonzero polynomial

in the x_i has a unique term of least value and cannot vanish. Hence $x_1, \dots, x_\ell$ are algebraically independent over k . Therefore

$$\ell \leq \text{trdeg}_k M.$$

This bounds the length of every prime chain in every valuation overring of T , and (6.1) follows. $\square$

Proof of Theorem 1.2. By Lemmas 3.1 to 3.3 and Proposition 4.1, the domain $D_{3,n}$ is local, has maximal ideal $\mathfrak{m}_{3,n}$, residue field k , and

$$\dim D_{3,n} = 1, \quad \dim D_{3,n}[X] = 3.$$

It remains to compute the second coordinate. Set

$$A_{3,n} = \text{Int}(D_{3,n}), \quad F = X^q - X.$$

By Proposition 5.1,

$$A_{3,n}[1/F] = C_{3,n}[X, 1/F]. \quad (6.2)$$

Put $B = C_{3,n}[X, 1/F]$. For $0 \leq i \leq n-2$, let

$$\mathfrak{p}_i = \mathfrak{q}_i B.$$

For each i there is a natural isomorphism

$$B/\mathfrak{p}_i \cong (C_{3,n}/\mathfrak{q}_i)[X, 1/\overline{F}],$$

where $\overline{F} = X^q - X$. Since $C_{3,n}/\mathfrak{q}_i$ is a domain and $\overline{F}$ is a nonzero monic polynomial, the ring on the right is a domain. Thus every $\mathfrak{p}_i$ is prime. Its constant subring is $C_{3,n}/\mathfrak{q}_i$, so the strict inclusions in (5.4) remain strict after extension and localization. Hence

$$(0) \subsetneq \mathfrak{p}_0 \subsetneq \mathfrak{p}_1 \subsetneq \dots \subsetneq \mathfrak{p}_{n-2} \quad (6.3)$$

is a chain of length $n-1$.

By Proposition 5.2,

$$B/\mathfrak{p}_{n-2} \cong k[X, 1/F].$$

Choose an irreducible factor G of the nonconstant polynomial $F - 1 = X^q - X - 1 \in k[X]$. Since $F \equiv 1 \pmod{G}$, localization at F does not kill (G) , and

$$k[X, 1/F]/(G)k[X, 1/F] \cong k[X]/(G)$$

is a field. Let $\mathfrak{p}_{n-1}$ be the inverse image of $(G)k[X, 1/F]$ in B . Then $\mathfrak{p}_{n-2} \subsetneq \mathfrak{p}_{n-1}$, so (6.3) extends to a prime chain of length n in B .

By (6.2), $B = A_{3,n}[1/F]$. The localization correspondence is an order isomorphism between $\text{Spec}(A_{3,n}[1/F])$ and the primes of $A_{3,n}$ disjoint from $\{1, F, F^2, \dots\}$. Contracting the chain to $A_{3,n}$ therefore preserves every strict inclusion, and

$$\dim A_{3,n} \geq n. \quad (6.4)$$

The inclusions

$$D_{3,n}[X] \subseteq A_{3,n} \subseteq \Omega[X]$$

and $\text{Frac}(D_{3,n}) = \Omega$ imply

$$\text{Frac}(A_{3,n}) = \Omega(X).$$

Indeed, the left inclusion gives $\Omega(X) = \text{Frac}(D_{3,n}[X]) \subseteq \text{Frac}(A_{3,n})$, while the right inclusion gives the reverse containment. The field Ω is algebraic over $L(t)$, because each $t^{1/\ell!}$ is algebraic over $L(t)$. Thus $\Omega(X)$ is algebraic over

$$k(u_1, \dots, u_{n-2}, t, X),$$

which is a rational function field of transcendence degree n over k . Therefore Lemma 6.1 gives $\dim A_{3,n} \leq n$. Together with (6.4), this proves $\dim \text{Int}(D_{3,n}) = n$. $\square$

7. REALIZING PAIRS ON AND ABOVE THE DIAGONAL BY SUSPENSION

The base-row theorem gives the decisive counterexamples and realizes every pair $(3, n)$ with $n \geq 3$. To pass from that family to the full classification, we now raise both dimensions in lockstep. We include the pullback argument in full because prime chains crossing the conductor require care.

7.1. A one-step suspension. We first prove the dimension formula for the pullback used to change the first coordinate of the dimension pair. The following elementary lemma is the point at which the hypothesis that the residue field is exactly a quotient field is important.

Lemma 7.1 (Pullback suspension). *Let A be a domain with quotient field K , let $A[X] \subseteq B \subseteq K[X]$ be a domain of finite dimension d , and let*

$$W = K[Y]_{(Y)}, \quad M = YW.$$

Write $\lambda : W[X] \rightarrow K[X]$ for coefficientwise reduction modulo M and put $B^\sharp = \lambda^{-1}(B)$. Then

$$\dim B^\sharp = d + 1.$$

Proof. Put $J = M[X]$, an ideal of both $W[X]$ and $B^\sharp$. We have

$$B^\sharp/J \cong B, \quad B^\sharp[1/Y] = K(Y)[X].$$

For the second equality, observe that $Yg \in J \subseteq B^\sharp$ for every $g \in W[X]$, and that $W[1/Y] = K(Y)$. Moreover, for a prime P of $B^\sharp$, one has

$$J \subseteq P \iff Y \in P.$$

Only the reverse implication requires comment. If $Y \in P$ and $h = Yg \in J$, then $h^2 = Y(Yg^2)$ with $Yg^2 \in J \subseteq B^\sharp$; hence $h^2 \in P$ and $h \in P$.

It follows that chains of primes not containing J —equivalently, chains whose primes avoid the powers of Y —are carried, with all strict inclusions preserved, to chains in the one-dimensional PID $K(Y)[X]$. Thus such a chain has at most one edge. Primes containing J correspond, in an order-preserving way, to primes of B .

We also need the following compatibility at the crossing. No nonzero prime P of $B^\sharp$ that does not contain J can be contained in J . Indeed, localize P in $K(Y)[X]$ and choose a nonzero polynomial in the resulting prime. After multiplying by an element of $K(Y)^\times$, it may be taken to be a polynomial $g \in W[X]$ having at least one unit coefficient. Its reduction $\bar{g} \in K[X]$ is therefore nonzero. Since $K = \text{Frac}(A)$, there is $0 \neq a \in A$ which clears the finitely many denominators of the coefficients of $\bar{g}$. Then

$$ag \in PB^\sharp[1/Y] \cap B^\sharp, \quad \lambda(ag) = a\bar{g} \in A[X] \subseteq B, \quad \lambda(ag) \neq 0.$$

Since $Y \notin P$, contraction from the localization gives

$$PB^\sharp[1/Y] \cap B^\sharp = P.$$

Thus $ag \in P \setminus J$, as claimed.

Consider now an arbitrary chain in $\text{Spec}(B^\sharp)$. If none of its primes contains J , it has length at most 1. If every prime contains J , its length is at most d . Otherwise let P be the last prime not containing J and P' the first prime containing J . If the initial block has no edge, the transition $P \subsetneq P'$ followed by a chain in B gives at most $1 + d$ edges. If that block has one edge, then $P \neq (0)$. The image P'/J is nonzero, for otherwise $P' = J$, contradicting the preceding paragraph. A chain in the domain B beginning at the nonzero prime P'/J has at most $d - 1$ further edges, since it can be preceded by the strict inclusion $(0) \subsetneq P'/J$. The total is again at most

$$1 + 1 + (d - 1) = d + 1.$$

Hence $\dim B^\sharp \leq d + 1$.

Conversely, choose a chain $(0) = \mathfrak{p}_0 \subsetneq \cdots \subsetneq \mathfrak{p}_d$ of length d in B . Its inverse images form a chain beginning with J , and prepending $(0) \subsetneq J$ gives a chain of length $d + 1$ in $B^\sharp$. $\square$

The integer-valued dimension formula in the following proposition is a special case of Tartarone's divided-prime pullback formula, recalled and generalized in [6, Section 1 and Theorem 1.2]. The corresponding coefficient-reduction map appears in [6, Lemma 1.1(2)]. We include a direct proof of the dimension formulas and the exact pullback identity used here.

Proposition 7.2 (One-step suspension). *Let $(A, \mathfrak{a})$ be a local domain with $0 \neq \mathfrak{a}$, quotient field K , and residue field $k = \mathbb{F}_q$. Let $W = K[Y]_{(Y)}$, let $\pi : W \rightarrow K$ be reduction modulo YW , and define*

$$A^\sharp = \pi^{-1}(A).$$

Then $A^\sharp$ is a local domain with residue field k and quotient field $K(Y)$, and

$$\dim A^\sharp = \dim A + 1, \tag{7.1}$$

$$\dim A^\sharp[X] = \dim A[X] + 1, \tag{7.2}$$

$$\dim \text{Int}(A^\sharp) = \dim \text{Int}(A) + 1. \tag{7.3}$$

Here the dimensions in the last two formulas are assumed finite.

Proof. The ring $A^\sharp$ is local with maximal ideal $\mathfrak{a}^\sharp = \pi^{-1}(\mathfrak{a})$: an element outside this ideal has residue a unit of A , and its inverse in W has residue in A as well. The residue field is therefore k . Also $YW \subseteq A^\sharp$, and multiplication by Y shows $W \subseteq \text{Frac}(A^\sharp)$; hence $\text{Frac}(A^\sharp) = K(Y)$.

Every nonzero prime of $A^\sharp$ contains $M = YW$. Indeed, let $0 \neq x$ belong to such a prime P . If $x \notin M$, then x is a unit of W , and $y/x \in M \subseteq A^\sharp$ for every $y \in M$; hence $y = x(y/x) \in P$. If $x \in M$, write $e = \text{ord}_Y(x) > 0$. For $0 \neq y \in M$, choose N with $N \text{ord}_Y(y) > e$. Then $y^N/x \in M \subseteq A^\sharp$, so $y^N \in P$ and therefore $y \in P$. Thus $M \subseteq P$ in both cases. Primes containing M are precisely the inverse images of primes of A . Consequently

$$\text{Spec}(A^\sharp) = \{(0)\} \cup \{\pi^{-1}(\mathfrak{p}) : \mathfrak{p} \in \text{Spec}(A)\},$$

and (7.1) follows.

We shall need one coefficient consequence. If $c \in \mathcal{C}(A^\sharp)$ has witness N , choose $0 \neq a \in \mathfrak{a}$ and a lift $y \in A^\sharp$ of a . Since $\pi(y) = a \neq 0$, the element y is a unit of W . The inclusion $cy^N \in A^\sharp \subseteq W$ therefore implies $c \in W$. Hence

$$\mathcal{C}(A^\sharp) \subseteq W. \tag{7.4}$$

Coefficientwise reduction identifies

$$A^\sharp[X] = \pi_X^{-1}(A[X]) \subseteq W[X].$$

Formula (7.2) is therefore Lemma 7.1 applied to $B = A[X]$.

For the integer-valued polynomial ring we first claim the exact pullback identity

$$\text{Int}(A^\sharp) = \pi_X^{-1}(\text{Int}(A)) \subseteq W[X]. \tag{7.5}$$

By the Vandermonde coefficient argument in Proposition 5.1, every coefficient of an element of $\text{Int}(A^\sharp)$ lies in $\mathcal{C}(A^\sharp)$, which is contained in W by (7.4). Such a polynomial can therefore be reduced coefficientwise. Evaluating it at arbitrary lifts of elements of A shows that its reduction is in $\text{Int}(A)$. This proves one inclusion in (7.5). Conversely, if $f \in W[X]$ and $\pi_X(f) \in \text{Int}(A)$, then, for every $x \in A^\sharp$,

$$\pi(f(x)) = \pi_X(f)(\pi(x)) \in A.$$

Thus $f(x) \in \pi^{-1}(A) = A^\sharp$, proving the reverse inclusion. Finally, $A[X] \subseteq \text{Int}(A) \subseteq K[X]$ and $\text{Frac}(\text{Int}(A)) = K(X)$, so applying Lemma 7.1 with $B = \text{Int}(A)$ to (7.5) gives (7.3). $\square$

7.2. Realization on and above the diagonal.

Theorem 7.3 (Realization on and above the diagonal). *Let k be a finite field, and let m, n be integers with*

$$3 \leq m \leq n.$$

There is a local domain $(D_{m,n}, \mathfrak{m}_{m,n})$ such that

$$D_{m,n}/\mathfrak{m}_{m,n} \cong k, \quad \dim D_{m,n} = m - 2,$$

and

$$\dim D_{m,n}[X] = m, \quad \dim \text{Int}(D_{m,n}) = n.$$

Proof. Since $n - m + 3 \geq 3$, Theorem 1.2 supplies the seed domain $D_{3,n-m+3}$. Apply the one-step suspension of Proposition 7.2 exactly $m - 3$ times, each time using a new indeterminate and the quotient field of the preceding domain as the residue field of the new discrete valuation ring. Call the result $D_{m,n}$; when $m = 3$, this means $D_{m,n} = D_{3,n}$. Every suspension raises the dimension of the domain by one, so

$$\dim D_{m,n} = 1 + (m - 3) = m - 2.$$

The same calculation for the other two dimensions gives

$$\begin{aligned} \dim D_{m,n}[X] &= 3 + (m - 3) = m, \\ \dim \text{Int}(D_{m,n}) &= (n - m + 3) + (m - 3) = n. \end{aligned}$$

The residue field remains k at every suspension. $\square$

Thus every pair with $3 \leq m \leq n$ is realized. To complete the classification, only Cahen's lower boundary $n = m - 1$ and the rigid cases $m = 1, 2$ remain.

8. THE LOWER BOUNDARY AND COMPLETION OF THE CLASSIFICATION

We now combine the preceding construction with the classical example on Cahen's lower boundary $n = m - 1$ and the two low-dimensional cases.

We begin with the classical pseudo-valuation-domain example recorded in [3, Example V.1.12] and [8, Equations (3.3)–(3.5), p. 454]. We prove its two dimension statements directly so that the classification remains self-contained.

Proposition 8.1 (Classical lower-bound example). *Let $k = \mathbb{F}_q$, let $L = k(U)$, and put*

$$V = L[Y]_{(Y)}, \quad M = YV, \quad D_{3,2} = k + M, \quad K = \text{Frac}(V) = L(Y).$$

Then

$$\dim D_{3,2}[X] = 3, \quad \dim \text{Int}(D_{3,2}) = 2.$$

Proof. Reduction modulo M identifies $D_{3,2}/M$ with k . If $a + z \notin M$, where $a \in k^\times$ and $z \in M$, then $a + z$ is a unit of V whose inverse has residue $a^{-1} \in k$; hence its inverse lies in $D_{3,2}$. Thus $D_{3,2}$ is local with maximal ideal M . Moreover, $Y \in D_{3,2}$ and $Yz \in M \subseteq D_{3,2}$ for every $z \in V$, so $V \subseteq \text{Frac}(D_{3,2})$ and therefore $\text{Frac}(D_{3,2}) = K$.

There is no nonzero prime strictly below M . Indeed, if P is a nonzero proper prime of $D_{3,2}$, choose $0 \neq x \in P \subseteq M$. For every $0 \neq y \in M$, choose N such that $N \text{ord}_Y(y) > \text{ord}_Y(x)$. Then $y^N/x \in M \subseteq D_{3,2}$, so $y^N \in P$ and $y \in P$. Hence $M \subseteq P$, and $P = M$. Consequently

$$\text{Spec}(D_{3,2}) = \{(0), M\}.$$

Let $\rho : V \rightarrow L$ be reduction modulo M . Apply Lemma 2.3 with

$$A = D_{3,2}, \quad \theta = U, \quad c = Y.$$

Indeed, $\ker(\rho|_{D_{3,2}}) = M$, $\rho(D_{3,2}) = k$, the element $\rho(U) = U$ is transcendental over k , and $YU \in M \subseteq D_{3,2}$. Thus $\dim D_{3,2}[X] = 3$.

Set $A_{3,2} = \text{Int}(D_{3,2})$ and

$$J = \{f \in A_{3,2} : f(D_{3,2}) \subseteq M\}.$$

Reduction after evaluation defines a homomorphism

$$\Phi : A_{3,2} \longrightarrow \prod_{a \in D_{3,2}} k, \quad f \longmapsto (f(a) + M)_{a \in D_{3,2}},$$

whose kernel is J . Hence J is radical. Plainly $MA_{3,2} \subseteq J$. Conversely, if $f \in J$, then $f(a) \in YV$ for every $a \in D_{3,2}$, and so

$$\frac{f(a)^2}{Y} \in YV = M \subseteq D_{3,2}.$$

Thus $f^2/Y \in A_{3,2}$, whence $f^2 \in YA_{3,2} \subseteq MA_{3,2}$. We have proved

$$\sqrt{MA_{3,2}} = J.$$

Every prime of $A_{3,2}$ contracting to M therefore contains J . Furthermore, each element z of $A_{3,2}/J$ satisfies $z^q = z$, because this identity holds in every factor of the displayed product. Every prime quotient of $A_{3,2}/J$ is then a field: in that quotient, each nonzero z satisfies $z^{q-1} = 1$. Consequently $A_{3,2}/J$ is zero-dimensional, and a prime chain of $A_{3,2}$ contracting throughout to M has no edge.

On the other hand,

$$(D_{3,2} \setminus \{0\})^{-1}A_{3,2} = K[X],$$

because $D_{3,2}[X] \subseteq A_{3,2} \subseteq K[X]$. Hence a prime-chain block contracting to (0) localizes, with all strict inclusions preserved, to a chain having at most one edge. Since these are the only two contractions, an arbitrary chain in $A_{3,2}$ has at most one such edge and one transition; thus $\dim A_{3,2} \leq 2$.

Finally, evaluation at zero is a surjection $A_{3,2} \rightarrow D_{3,2}$. Its kernel Q_0 is a nonzero prime because $X \in Q_0$, and the inverse image P_0 of M is a prime properly containing Q_0 , since the constant Y belongs to $P_0 \setminus Q_0$. Therefore

$$(0) \subsetneq Q_0 \subsetneq P_0$$

is a length-two chain. This proves $\dim \text{Int}(D_{3,2}) = 2$. $\square$

Proof of Theorem 1.1. Suppose first that a domain D realizes (m, n) . If $m = 1$ or $m = 2$, Proposition 2.2 gives respectively $(m, n) = (1, 1)$ or $(m, n) = (2, 2)$. If $m \geq 3$, Cahen's lower bound in Proposition 2.1 gives $n \geq m - 1$. This proves the necessity of condition (ii).

Conversely, any field may serve as $D_{1,1}$; then $\text{Int}(D_{1,1}) = D_{1,1}[X]$, and both rings have dimension 1. For the pair $(2, 2)$, take an algebraically closed field k and put $D_{2,2} = k[T]$. We show that $\text{Int}(D_{2,2}) = D_{2,2}[X]$. Write an integer-valued polynomial as $f = g/d$, where $0 \neq d \in k[T]$, $g \in k[T][X]$, and no irreducible factor of d divides every coefficient of g . If an irreducible π divides d , then $g(a) \in dk[T] \subseteq \pi k[T]$ for every $a \in k[T]$. Because k is algebraically closed, $k[T]/(\pi) \cong k$. Reduction modulo π would therefore make the nonzero polynomial $\bar{g} \in k[X]$ vanish at every element of the infinite field k , a contradiction. Thus d is a unit, so $\text{Int}(D_{2,2}) = D_{2,2}[X]$, and both rings have dimension 2.

Let $m \geq 3$, and fix any finite field k . If $n \geq m$, Theorem 7.3 realizes the pair by a local domain $D_{m,n}$ with residue field k . It remains only to realize $n = m - 1$. Start with the domain $D_{3,2}$ of Proposition 8.1, formed over this k , and apply the one-step suspension of Proposition 7.2 exactly $m - 3$ times. Call the result $D_{m,m-1}$; for $m = 3$, this means the original $D_{3,2}$. Each step preserves locality and the residue field and adds one to all three relevant dimensions, so

$$\dim D_{m,m-1} = 1 + (m - 3) = m - 2,$$

and

$$\dim D_{m,m-1}[X] = 3 + (m - 3) = m, \quad \dim \text{Int}(D_{m,m-1}) = 2 + (m - 3) = m - 1.$$

Hence every pair in (ii) occurs. $\square$

CONCLUSION

Problem 14 therefore has a negative answer. The domain $D_{3,4}$ already satisfies

$$\dim D_{3,4}[X] = 3 < 4 = \dim \operatorname{Int}(D_{3,4}),$$

but the phenomenon is much stronger than a single counterexample: among one-dimensional local domains with finite residue field and with $\dim D[X] = 3$, the value of $\dim \operatorname{Int}(D)$ is unbounded.

At the level of finite dimension pairs, the classification is complete. The pairs $(1, 1)$ and $(2, 2)$ are the only possibilities with first coordinate at most 2, whereas for every $m \geq 3$ the possible second coordinates are exactly the integers $n \geq m - 1$. Every admissible pair with $m \geq 3$ is realized by a local domain $D_{m,n}$ with prescribed finite residue field and $\dim D_{m,n} = m - 2$. Thus there is no finite upper bound for $\dim \operatorname{Int}(D)$ in terms of $\dim D[X]$ alone, while Cahen's lower bound remains sharp and is the sole numerical restriction once $\dim D[X] \geq 3$.

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DEPARTMENT OF MATHEMATICS, NATIONAL UNIVERSITY OF SINGAPORE, SINGAPORE 119076

Email address: `hoang.tranviet@u.nus.edu`

URL: `https://vh-tran.github.io/`

DEPARTMENT OF MATHEMATICS, NATIONAL UNIVERSITY OF SINGAPORE, SINGAPORE 119076

Email address: `tanmn@nus.edu.sg`

URL: `https://tanmnguyen89.github.io/`